STRAVA FITNESS

Power BI Report

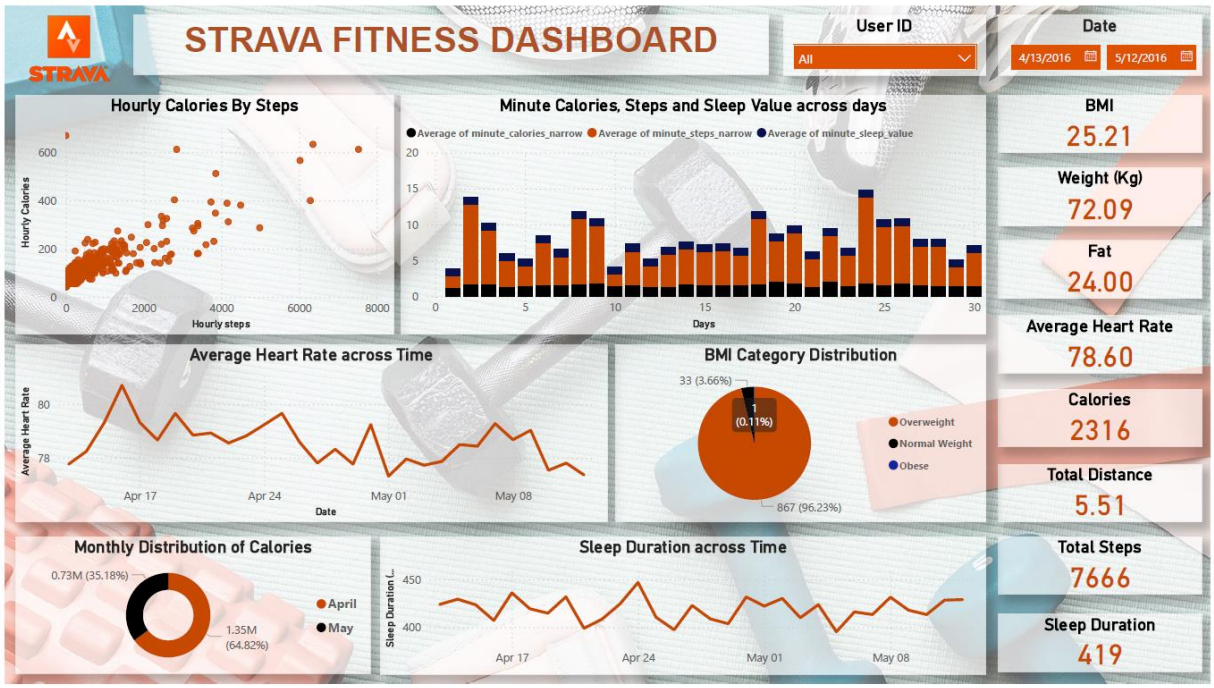
INTRODUCTION

This Power BI dashboard analyzes Strava fitness data for a one-month period (April 13 to May 12, 2016). It provides a holistic view of user health by integrating activity metrics, physiological data (BMI, weight, heart rate), and sleep patterns to understand overall wellness trends.

OBJECTIVE

The primary goal of this dashboard is to visualize the correlation between daily activities (steps, calories) and health outcomes (BMI, heart rate, sleep). It aims to identify patterns in user behavior and physiological health to provide actionable fitness insights.

STRAVA FITNESS DASHBOARD



DAX MEASURES USED

Based on the visual elements, the following core DAX measures were utilized:

- **Average Metrics:** Average Heart Rate (183), BMI (29.33), and Weight (40.21).
- **Total Aggregations:** Total Distance (10.06km), Total Steps (41,709K), and Total Minutes Calories (247K).

- **Percentage of Total:** Used in the BMI Category Distribution and Monthly Distribution of Calories.

KEY PERFORMANCE INDICATORS (KPIs)

The dashboard prominently tracks several health and fitness KPIs:

- **BMI:** 25.21 (Overweight)
- **Weight (Kg):** 72.09
- **Fat:** 24.00
- **Average Heart Rate:** 78.60 BPM
- **Calories:** 2,316
- **Total Distance:** 5.51
- **Total Steps:** 7,666
- **Sleep Duration:** 419 Minutes (~7 hours)

DATA VISUALIZATIONS

- **Hourly Calories By Steps (Scatter Plot):** Displays a strong positive correlation between hourly steps and calories burned. Most data points cluster between 0–2,000 steps, while high-intensity outliers reach over 600 calories.
- **Minute Calories, Steps and Sleep Value (Stacked Column Chart):** Tracks the daily volume of activity across 30 days. It reveals fluctuations in daily consistency, showing peaks and troughs in calorie expenditure and step counts.
- **Average Heart Rate across Time (Line Chart):** Monitors the daily average heart rate, showing a range between ~77 and ~80 BPM. This indicates a relatively stable resting/low-activity heart rate over the month.
- **BMI Category Distribution (Pie Chart):** A critical health metric showing that 96.23% (867 instances) of the data falls into the Overweight category, with only 3.66% (33 instances) in the Normal Weight category.
- **Monthly Distribution of Calories (Donut Chart):** Breaks down activity by month. April (64.82% / 1.35M calories) saw nearly double the engagement and calorie burn compared to May (35.18% / 0.73M calories).
- **Sleep Duration across Time (Line Chart):** Visualizes sleep patterns over the 30-day period. It shows significant variability, with the average duration sitting at 419 minutes (approximately 7 hours).

SLICERS

To allow for interactive data exploration, the dashboard includes:

- **Date Range Slicer:** Set from 4/13/2016 to 5/12/2016.
- **ID Slicer:** Allows for filtering the entire report by specific individual participants.

BUSINESS INSIGHTS SUMMARY

- **Health Profile:** The overwhelming majority (**96.23%**) of users are in the "Overweight" BMI category (25.21), suggesting a target market for weight-loss-focused features.
- **Activity Gap:** The **Total Steps (7,666)** fall short of the universally recommended 10,000-step goal for optimal health.
- **Retention Drop:** There is a significant decline in activity volume from **April to May**, dropping from 64.82% to 35.18% of total monthly calories.

Business Recommendations

- **Personalized Coaching:** Introduce a "Road to 10k" step challenge to help users bridge the gap from 7,666 to 10,000 steps.
- **Weight Management Tools:** Since BMI is high (25.21), integrate a calorie-deficit tracker or nutritional logging to complement the physical activity data.
- **Engagement Re-ignition:** Implement push notifications or mid-month "milestone" rewards to prevent the type of engagement drop seen in May.

BUSINESS RECOMMENDATIONS

- **Targeted Weight Management:** Develop specialized "Step Challenges" for users in the higher BMI categories to encourage gradual weight loss.
- **Heart Rate Monitoring:** Integrate "Recovery Zone" alerts for users maintaining high average heart rates to ensure they are resting sufficiently.
- **Sleep Hygiene:** Given the variability in sleep duration, introduce in-app prompts for consistent sleep schedules to improve overall recovery.

CONCLUSION

The Strava Fitness Dashboard effectively highlights that while users are achieving high step counts and calorie burns, their physiological markers (BMI and Heart Rate) suggest a need

for more balanced health coaching. By focusing on consistency and recovery, the business can move beyond simple tracking to becoming a true health improvement partner. The dashboard reveals a user base that is moderately active but faces specific health risks related to BMI. By leveraging the clear correlation between steps and calories shown in the scatter plot, Strava can drive better health outcomes through increased daily movement targets and consistent sleep tracking.